Module 4 — Showers, MPI, and Tunes

Supplementary to the sPHENIX Standalone Course

1. Why parton showers matter

A 2→2 matrix element gives you two hard partons. Real collisions have hundreds of particles. The shower is the step that bridges this gap: repeated splittings — q→qg, g→gg, g→qq■ — generated down to a cutoff of around 1 GeV, where hadronization takes over.

2. DGLAP evolution

The shower is a Markov chain in virtuality (or p_T). At each step, the splitting probability is $dP = (\alpha_s/2\pi) \cdot P(z) \cdot dQ^2/Q^2\, dz$, where $P(z)$ is the Altarelli-Parisi splitting function. Sudakov factors enforce ordering in Q^2 so the probability is properly normalized.

Splitting	Splitting function P(z)
q → qg	$(4/3) \cdot (1+z^2) / (1-z)$
g → gg	$3 \cdot [z/(1-z) + (1-z)/z + z(1-z)]$
g → qq■	$(1/2) \cdot [z^2 + (1-z)^2]$

3. ISR vs FSR

Pythia splits the shower into Initial State Radiation (off the incoming partons) and Final State Radiation (off the outgoing ones). You can toggle each:

```
PartonLevel:ISR = on
PartonLevel:FSR = on
// Per-cutoff controls:
TimeShower:pTmin = 0.5   // FSR cutoff [GeV]
SpaceShower:pT0Ref = 2.0 // ISR reference
```

Turning off a shower is a useful diagnostic: if your observable changes dramatically, that observable is shower-dominated.

4. Multi-parton interactions

Multiple parton interactions (MPI) occur when more than one pair of partons from the two incoming protons scatter. MPI is the dominant source of 'underlying event' — the ambient activity that isn't from the hard process.

```
PartonLevel:MPI = on
MultipartonInteractions:pT0Ref = 2.28   // GeV, Monash default
MultipartonInteractions:ecmRef = 7000.  // reference √s
MultipartonInteractions:ecmPow = 0.215  // energy scaling
```

5. Colour reconnection

After the partonic cascade, soft gluons between different parton pairs can 'reconnect' colour strings before hadronization. This affects baryon-to-meson ratios and particle correlations.

```
ColourReconnection:reconnect = on
ColourReconnection:mode = 0     // MPI-based (default)
```

6. Lund string fragmentation

Partons are confined: a q and q■ moving apart can be modelled as a string of uniform tension (~1 GeV/fm). The string breaks into hadrons via q-q■ pair production in its field, with a Gaussian distribution in p_T and an exponential distribution in longitudinal momentum fraction z (the 'Lund function').

```
StringPT:sigma = 0.335        // Gaussian width in pT [GeV]
StringZ:aLund = 0.68          // z distribution shape
StringZ:bLund = 0.98
```

7. Tunes

Because showers+MPI+hadronization have many parameters, physicists produce 'tunes' — coherent parameter sets that describe data. Each LHC experiment publishes its own. For generic use, the **Monash 2013** tune (Tune:pp = 14) is the default and widely trusted.

	Name	Notes
	Pythia 8.200 default	Older, replaced by Monash
	Monash 2013	Current default, good at LHC and RHIC energies
	ATLAS A14 CTEQ6L1	ATLAS Run-2 central tune
	CMS CP5	CMS Run-2 central tune
	Various ATLAS AZ/AU2/A14 variants	ISR/FSR/MPI up/down variations

8. Varying showers for systematics

For a systematic uncertainty on your observable from shower modelling, you typically run two variations:

```
# weak-shower variation: FSR off
PartonLevel:FSR = off
# strong-shower variation: tight cutoff
```

```
TimeShower:pTmin = 1.0
```

Take the envelope of the two variations, symmetrised, as the FSR systematic. Do the same for ISR.

9. Practical checks

Check	What it tells you
Mean charged mult vs √s	MPI strength and ISR tuning
Jet multiplicity in Z+jets	ISR matches tune
Underlying event transverse region	MPI p_T0 is right
Inclusive jet p_T spectrum	Hard process × PDFs × shower joint
Charged-particle p_T spectrum	Tune validity across kinematic range

10. Diagnostic plot recipe

A shower/MPI study is easiest to interpret if you hold one variable constant and vary another. Pick one observable (e.g. charged multiplicity at $|\eta|<0.5$), then generate the same sample with five different tune IDs:

```
for tune in 7 14 18 22 32; do
    mkdir -p out/tune$tune
    cd out/tune$tune
    cat > cmnd.txt <<EOF
Beams:eCM = 200.
SoftQCD:nonDiffractive = on
Tune:pp = $tune
Random:setSeed = on
Random:seed = 100
EOF
    ../../bin/mypythia cmnd.txt > run.log
    cd ../..
done
```

Overlay the resulting histograms and note which tune best matches data. The 'best' tune varies by observable and energy — that's why there are so many.

11. What to vary for a systematic

- **ISR variation:** SpaceShower:pT0Ref × 2 and / 2
- **FSR variation:** TimeShower:alphaSvalue ± 10%
- **MPI variation:** MultipartonInteractions:pT0Ref ± 10%
- **Hadronization:** StringZ:aLund, StringZ:bLund variations
- **Colour reconnection:** ColourReconnection:mode 0, 1, 2

Quote the envelope of these as a shower-modelling systematic. This is what ATLAS and CMS do; it's also good practice for a standalone analysis.